

Assignment – 2
Quantum Mechanics – II
Assignment due on March 26

1.

Prove

$$\sigma_{\text{tot}} \simeq \frac{m^2}{\pi \hbar^4} \int d^3x \int d^3x' V(r) V(r') \frac{\sin^2 k|\mathbf{x} - \mathbf{x}'|}{k^2 |\mathbf{x} - \mathbf{x}'|^2}$$

in each of the following ways.

- a. By integrating the differential cross section computed using the first-order Born approximation.
- b. By applying the optical theorem to the forward-scattering amplitude in the *second-order* Born approximation. [Note that $f(0)$ is real if the first-order Born approximation is used.]

2.

Consider a potential

$$V = 0 \quad \text{for } r > R, \quad V = V_0 = \text{constant} \quad \text{for } r < R,$$

where V_0 may be positive or negative. Using the method of partial waves, show that for $|V_0| \ll E = \hbar^2 k^2 / 2m$ and $kR \ll 1$ the differential cross section is isotropic and that the total cross section is given by

$$\sigma_{\text{tot}} = \left(\frac{16\pi}{9} \right) \frac{m^2 V_0^2 R^6}{\hbar^4}.$$

Suppose the energy is raised slightly. Show that the angular distribution can then be written as

$$\frac{d\sigma}{d\Omega} = A + B \cos \theta.$$

Obtain an approximate expression for B/A .

3. Prove the Optical Theorem

4.

Consider the scattering of a particle by an impenetrable sphere

$$V(r) = \begin{cases} 0 & \text{for } r > a \\ \infty & \text{for } r < a. \end{cases}$$

- Derive an expression for the s -wave ($l = 0$) phase shift. (You need not know the detailed properties of the spherical Bessel functions to be able to do this simple problem!)
- What is the total cross section $\sigma [\sigma = \int (d\sigma/d\Omega) d\Omega]$ in the extreme low-energy limit $k \rightarrow 0$? Compare your answer with the geometric cross section πa^2 . You may assume without proof:

$$\frac{d\sigma}{d\Omega} = |f(\theta)|^2,$$

$$f(\theta) = \left(\frac{1}{k}\right) \sum_{l=0}^{\infty} (2l+1) e^{i\delta_l} \sin \delta_l P_l(\cos \theta).$$

5.

Use $\delta_l = \Delta(b)|_{b=l/k}$ to obtain the phase shift δ_l for scattering at high energies by (a) the Gaussian potential, $V = V_0 \exp(-r^2/a^2)$, and (b) the Yukawa potential, $V = V_0 \exp(-\mu r)/\mu r$. Verify the assertion that δ_l goes to zero very rapidly with increasing l (k fixed) for $l \gg kR$, where R is the “range” of the potential. [The formula for $\Delta(b)$ is given in (6.168).] It is useful to realize that

$$\int_1^{\infty} \frac{e^{-ax}}{(x^2 - 1)^{1/2}} dx = K_0(x)$$

where $K_0(x)$ is the modified Bessel function of the second kind, of order zero.